

Shen (Sam) Wang

Boston, MA • 781-692-7014

wang.shen3@northeastern.edu • www.linkedin.com/in/ShenWang19840222 • Green Card

Education

M.S. in Software Engineering Systems (GPA 3.7), Northeastern University, Boston, MA Dec 2024

Relevant Courses: Concept of Object Oriented Design, Program Structure and Algorithm, Operating Systems, Web design UI/UX Engineering, Network Structure and Cloud Computing

Ph.D in Particle Physics and Nuclear Physics, Chinese Academy of Sciences, Shanghai, China Jul 2013

Technical Skills

Programming Languages:	C, Java, Python, JavaScript, Tcl/tk
Backend framework:	FastAPI, Flask
Relational Database and libraries:	MySQL, SQLite, Ssqlalchemy
Frontend UI framework:	HTML, CSS, Node.js, React.js
Machine learning and Data Science tools:	PyMC, Scikit-learn, Dask, Keras, PyTorch, Huggingface
Tools and platform:	Linux, VIM, Vscode, Git, IntelliJ IDEA

Academic Projects

Web Application Deployment Dec 2024

- Designed and implemented a RESTful API using FastAPI, ensuring efficient data handling
- Utilized Terraform to deploy the application on AWS, automating the configuration of infrastructure resources
- Configured DNS, RDS, S3, CloudWatch, Auto-Scaling, and Load Balancer to ensure high availability
- Successfully integrated the application with AWS services to achieve a fully automated and scalable solution
- Outcome: 99.99% API uptime guaranteed with fully automated CI/CD pipeline

A User-Friendly AI for Visualizing Skin Rashes Apr 2024

- Selected and tagged dermatological data of 3 human skin conditions: Eczema, Measles, Bowen' disease.
- Fine-tuned Stable Diffusion Model with Low Rank Adaptation for generating sample images.
- Built user friendly web based interface for generating dermatological images.
- Outcome: Enhanced diagnostic support by generating realistic dermatological images for early disease detection

Restaurant management system Apr 2023

- Created robust and reusable OOP code structures in Java, formed the backbone of the system.
- Developed a comprehensive User Interface using Java and JavaFX.
- Worked closely with a database specialist to ensure seamless integration with the backend database.

Professional Experiences

Boston College, Computer Science Dept, Boston, MA Apr 2024 - Sept 2024

- Development and Maintenance of the pytc-client Library for PyTorch Connectomics
- improved application stability by resolving critical inference disruptions and reduce errors by 30%
- Improved data analysis user experience by enabling multi-viewer visualization
- Increased accessibility for Windows developers by implementing windows specific scripts

Georgian Technical University (Top 3 in Georgia), Tbilisi, Republic of Georgia Jan 2019 - Jul 2022

Invited Professor in Biomedical Engineering

- Developed course “Medical Image Analysis” with C++ library ITK (Insight Tool Kit) for graduate students.

Izmir High Technology Institute (IYTE), Izmir, Turkey Jan 2016 - Dec 2016

Postdoctoral Researcher in Physics

- Implemented large-scale parallel simulations using Molecular Dynamics (MD) Software LAMMPS.
- Analyzed simulation data with C and tcl/tk to study heat conductivity of MoS₂ monolayer.